

9 (a) Define the *volt*.

.....
 [1]

(b) By reference to energy transfers, distinguish between e.m.f. and p.d.

e.m.f.

 p.d.
 [2]

(c) The current flowing through a light bulb can be controlled by a potential divider as shown in Fig. 9.1.

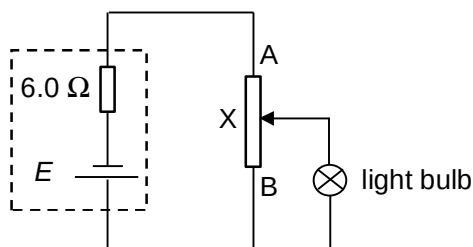


Fig. 9.1

AB is a potentiometer wire of uniform cross-sectional area with a total resistance of $16\ \Omega$. It is connected to a battery of e.m.f. E and internal resistance $6.0\ \Omega$ and a light bulb. When the slider of the potentiometer wire is set at X, exactly mid-way along AB, the bulb has an operating power of $0.50\ \text{W}$ and voltage of $2.0\ \text{V}$.

(i) Calculate the current through section XB of the potentiometer wire.

current = A [2]

(ii) Calculate the potential difference across section AX of the potentiometer wire.

potential difference = V [3]

(iii) Determine the e.m.f. E of the battery.

E = V [2]

(d) The current flowing through the light bulb can also be controlled using a variable resistor as shown in Fig. 9.2.

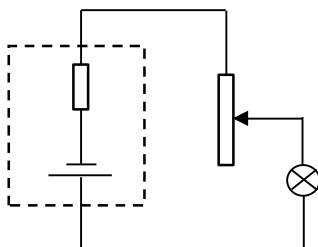


Fig. 9.2

State an advantage and disadvantage of using the variable resistor as compared to the potential divider in controlling the current through the light bulb.

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 [2]

- (e) Fig. 9.3 shows two cells E_1 and E_2 of negligible internal resistances connected with a variable resistor R in a potentiometer circuit. AB is a uniform resistance wire.

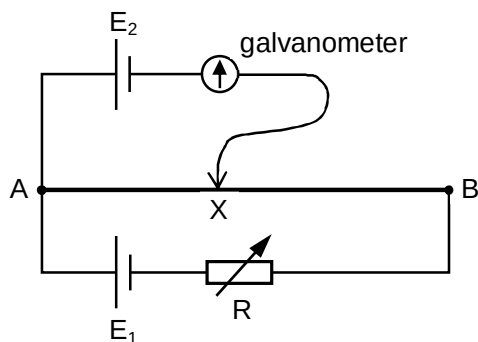


Fig. 9.3

- (i) When the sliding contact is moved from A towards B, explain why a point X is found where the galvanometer gives zero deflection.

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 [2]

- (ii) If the e.m.f. of E_1 is much larger than that of E_2 , explain why this circuit is not suitable for comparison of the e.m.f. of the cells.

.....

 [2]

- (f) Fig. 9.4 shows a potentiometer circuit with two cells E_1 and E_2 of negligible internal resistances for measuring a small e.m.f. produced by a thermocouple. Wire CD is 1.0 m long with a resistance of $2.0\ \Omega$. Cell E_2 has an e.m.f. of $1.1\ \text{V}$ and resistor R_1 has a resistance of $110\ \Omega$. Both galvanometers show no deflection.

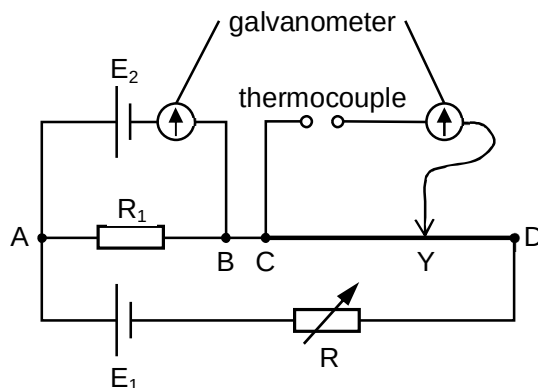


Fig. 9.4

- (i) State the potential difference across AB.

potential difference = V [1]

- (ii) The balance length CY is 72.5 cm.

Determine the e.m.f. of the thermocouple.

e.m.f. = V [3]

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